Notes

TotemLabs Compass:
https://totemlabs.com/blogs/posts/unity-mesh-network
https://totemlabs.com/blogs/posts/how-the-totem-compass-works
https://www.u-blox.com/en/totem-and-u-blox-navigate-the-future-of-human-connection

- Peer to peer mesh network
- Advertised max range between 2 devices 1000 meters
- Range increase with “unity mesh network” UMN
- Totem uses GNSS + UMN
- UMN uses data routing - allows nodes to choose which paths to take based on changing variables like speed/efficiency/security, so broken connections don’t break network
- Claims “boundless range” if 1-2% of attendees are users
- “Self healing algorithms” to account for dead zones, blocked/broken paths
- GNSS -> finds own location (unusable indoors for now)
- Connectionless peer to peer network that enables 2 way communication between devices
- Operate on 2.4GHz freq but on a lower latency protocol - 3 layers compared to wifi’s 6
- Uses IMU to switch into “compass mode” when laid flat
- Has SOS button
- Points to all different people “bonded” with using the various colors to indicate position

- Uses MAX-M10S GNSS module for positioning
- Claims it excels even in env. with significant interference (i.e festivals)
- Satellite connection between 30-150s

Reviews:

- https://www.reddit.com/r/aves/comments/1p3dsi8/totem_labs_review_awful_experience_with_a/
- Lotta negative
- https://www.wired.com/review/totem-labs-compass/
- Tricky setup
- Each compass pairs with up to 4 other ones
- Manageable via app, assign colors to each person

Crowd Compass:

https://www.crowdcompass.io/

- 915MHz radio + mesh network
    - High freq absorbed by water
    - Lower freq penetrates better
    - Uses LoRa protocol
- Send/recieve messages via phone app when connected to compass
- “Maplike” interface, shows each friend as a colored dot on device relative to your location
- Pairs with unlimited devices
- https://www.reddit.com/r/aves/comments/1gbh9u7/a_crowd_compass_review/
- Some reviews say accurate within ~20ft
- Sometimes need re-calibration

Apple:

- Find My Network: doesn’t work without wifi/cell connection, useless @ festivals
- Precision Finding: Direct connection via UWB either phone to phone or phone to airtag, WILL work at festivals
    - Caveat is that UWB requires you to be close already, roughly 10-30 meters

Meshtastic:

- Uses LoRa and nodes to transmit
- Mesh nodes ONLY respond to messages on the same network/code (like radio freqs)
- Channels sit above meshes, which have names and encryption keys
- A node can belong to 8 channel in the same mesh (all nodes in mesh will transmit/receive regardless of channel)
- LoRa different from FM/AM/PSK
- Uses chirp spread spectrum
    - Chirp is signal that changes frequency, up or down (upchirp, downchirp)
    - Chirp that starts at bottom and slides to top could be 0, but we can modify by for example starting the chirp in the middle, and ends elsewhere, it could symbolize a different value
    - Receiver will know how much a chirp is shifted by, and can therefore determine the right value
    - Spreading factor determines how many possible symbols
    - SF7 means each chirp can be one of 2^7=128 possible symbols, SF8 means 2^8 one of 4096 possible symbols: each chirp = 12bits of info
    - Catch to higher spread is each chirp takes longer, i.e 1ms vs 30ms
    - At the same time, higher SF means more range/reliability, just slower
        - Think of like whispering a word quickly vs dragging out that word

    - For tiny position updates (latitude, longitude, ID) ~16-20 bytes, use SF9-11 for more range/reliability
- Bandwidth
    - Bandwidth = how wide of frequency chirp sweeps
    - Wider = faster, more susceptible to interference
    - For long range outdoor, 125k-250kHz is typical
    - Combining SF + Bandwidth is what gives data rate, SF7+500k might be ~22kbps, SF12+125k might be 250bps
- Packets
    - Each LoRa signal is sent as a “packet”, a sequence of chirps in order
    - 1. Preamble
        - 16 standard unchanged chirps, allowing receiver to detect it and calibrate
    - 2. Sync Word
        - Pattern of chirps for identification, for example if 2 LoRa signals are on the same frequency at the same place, allows receiver to discern whether to accept or ignore it
    - 3. LoRa header
        - Metadata about the following packet: payload length, coding rate, CRC (?)
    - 4. Payload
        - The actual data being sent: who sent, who receives, type, actual message, etc, its encrypted
    - 5. CRC
        - Cyclic redundancy check
        - Math operation that hashes payload into a fingerprint
        - Sender sends a fingerprint, the operation is done and the receiver confirms the fingerprints match, if yes, pass on, if no, discard
- Forward Error Correction
    - Transmit extra data (redundant) to combat corruption of chirps
    - Allows receiver to mathematically reconstruct lost data
    - Coding Rate is what controls how MUCH redundancy
    - I.e coding rate ⅘ means every 4 bits of real data, add one bit of redundant
    - More redundancy = more reliability, but slower
    - Coding rate is in the header, so receiver can work with different rates

Radio Preset	Alt Preset Name	Data-Rate	SF / Symbols	Coding Rate	Bandwidth	Link Budget
Short Range / Turbo	Short Turbo	21.88 kbps	7 / 128	4/5	500 kHz[1]	140dB
Short Range / Fast	Short Fast	10.94 kbps	7 / 128	4/5	250 kHz	143dB
Short Range / Slow	Short Slow	6.25 kbps	8 / 256	4/5	250 kHz	145.5dB
Medium Range / Fast	Medium Fast	3.52 kbps	9 / 512	4/5	250 kHz	148dB
Medium Range / Slow	Medium Slow	1.95 kbps	10 / 1024	4/5	250 kHz	150.5dB
Long Range / Turbo	Long Turbo	1.34 kbps	11 / 2048	4/8	500 kHz	150dB
Long Range / Fast	Long Fast	1.07 kbps	11 / 2048	4/5	250 kHz	153dB
Long Range / Moderate	Long Moderate	0.34 kbps	11 / 2048	4/8	125 kHz	156dB
Long Range / Slow (depr.)	Long Slow	0.18 kbps	12 / 4096	4/8	125 kHz	158.5dB

Long moderate /Long Fast Should be good enough? (from meshtastic)
A position packet should take roughly 15 bytes? 120 bits
Link Budget:

- How much signal loss the radio can tolerate between sender before link breaks
- Higher dB = longer effective range
- Every 6 dB can roughly double range
- Festival crowd can cost 15-25dB compared to LoS

Comparison of Long Fast / Long Moderate
LF: 1.07kbps, each packet will need roughly 110ms on air / at 3packets/sec (account for rebroadcast due to packet flooding) 330ms of channel use per second - lots of headroom

LM: 0.34kbps, each packet roughly 350ms on air / at 3packets/sec = 1050ms, more than 1s per update = constant collisions and full/saturated channel, updates maybe ever 30+ seconds… too slow

LF > LM, even though 40% range tradeoff

Mesh Routing Protocol (Meshtastic)

- Problem with flooding is all devices are equal/weighted the same
    - Not efficient
- Meshtastic 2.0 algorithm gives preference to devices on edge of network
    - Gets it out further
    - More likely to hit the target receiver
    - Network prefers routers before going anywhere else (not really applicable i think)
- Not random delays between broadcasts, uses SNR Based Delay, delay is biased so that weaker signals rebroadcast first, stronger signals wait longer (see above)
-

Mesh Flooding

- Complete flooding does not work
- Need 3 rules for flooding
    - Hop Limit: each time a packet is rebroadcast, its limit -1, when it hits 0 it dies
    - Duplicate Suppression: each packet has an ID, when a device receives it checks if it recently received it, if it did it kills it, if not it will rebroadcast (hop limit permitting)
    - Delays between rebroadcast: prevents collisions of same packet,
- These 3 create *managed flooding*, what meshtastic uses

GNSS (Global Navigation Satellite System)

- Totem uses u-blox MAX-M10S
    - ~8-12$
    - Multi constellation, GPS GLONASS Galileo BeiDou, provides better accuracy and faster location
    - Very low power, 25mA acquisition, 12mA tracking, allows aggressive duty cycling
    - Supports AssistNow (u-blox's A-GNSS service) — pre-load satellite data over USB or BLE before festival, cold start time drops a lot
    - Documentation, community examples (lots), supported in many ecosystems
    - 10x10mm package
- ATGM336H
    - 3-5$
    - GPS & BeiDou only
    - Higher power consumption (35-45mA tracking)
    - Less mature A-GNSS support
    - Less community support

Probably just use uBlox not worth tradeoff for saving a few bucks

CPU Selection

Nordic nRF52840:

- Low power BLE “industry standard”
- Power management is good, deep sleep at single digit microamps
- SPI, UART, I2C, PWM, ADC
- 1mb ram, 256kb ram
- Onboard bluetooth
- ~4-6$

ESP32-S3

- Cheap 2-3$
- Lots of community support
- Bad power consumption (2-5x more active current draw than nordic, unstable deep sleep)

STM32L4

- Excellent power management, comparable to nordic
- No integrated BLE…
- Smaller community
- More learning curve

Add a feature so that it doesn’t transmit when the user hasn’t moved significantly (5 meters?) for 5 minutes or something, other devices will use its last known location unless the device starts moving again

Good for power saving?

- Not as much power saving as i thought, maybe around 5%, the LoRa only avgs 1.02mA, even if 80% less transmissions then only saves around 0.8mA
- Good for channel congestion though
    - So instead, maybe transmit position every 60s instead of 10s when stationary, cuts TX rate by 6x, and then if moving then go back to 10s

Power Budget Spreadsheet

https://docs.google.com/spreadsheets/d/1Pnn3CcJiRLiUpCU_vPH4JlqYkt9eogNWPeIxcUIkMIE/edit?usp=sharing

LED UX Design

- Direction, proximity, states (bonding, SOS, low batt…)
- 8 LEDs, 45 deg resolution
- How to show direction?
- How to layer distance without cluttering
- Distinguishable states
- READABLE

Design Info

Executive Summary

At large music festivals, cellular networks saturate and fail within the first hour, often for the rest of the event. Groups that get separated have no reliable way to find each other — the exact moment navigation is most needed is the moment phones stop working. Kandi is a wrist-worn device that solves this by creating its own offline network: group members' devices locate each other directly, with no cell service, Wi-Fi, or internet dependency, and each device guides its wearer toward the others through a simple ring of directional LEDs.

The problem is not unaddressed, but existing solutions each make a compromise Kandi avoids. The Totem Compass operates on 2.4GHz — the most congested band at a festival and one heavily absorbed by crowds of bodies — and is a bulky neck-worn pendant culturally out of place at a rave. The Crowd Compass made the better radio choice (sub-GHz LoRa, which penetrates bodies and avoids the 2.4GHz roar) but requires the user to pull out a phone app to read its map. Apple's offline finding works only at very short range and cannot bridge a festival. Kandi's thesis is to combine the right radio (sub-GHz LoRa mesh) with a screenless, glance-readable wrist interface, in a form factor built around kandi culture rather than against it — a beaded, LED-lit bracelet that reads as festival wear, not a gadget.

The technical architecture centers on a Semtech SX1262 LoRa radio at 915MHz, configured for ~500–800m of effective range in dense crowds, carrying tiny position updates across a managed-flooding mesh that supports groups of up to eight and uses per-group encryption keys so many groups can share the same frequency without interference. Positioning comes from a u-blox MAX-M10S multi-constellation GNSS module, with assisted-GNSS data preloaded before the event to eliminate slow cold-start times. A Nordic nRF52840 microcontroller anchors the system, providing both low-power operation and the Bluetooth radio used for pre-event bonding. The interface is an eight-LED directional ring that encodes direction by position, distance by blink rate, and member identity by color, controlled entirely by two buttons through layered gestures — no screen, readable at a glance even under strobes.

Power analysis projects a realistic 13–16 hours of operation on a small 400mAh cell, sufficient for a full festival day, aided by an adaptive duty-cycling strategy keyed on GPS-derived movement rather than raw motion — a deliberate choice so that dancing in place does not defeat the device's power savings the way a naïve motion trigger would. Close-range precision positioning via UWB was evaluated and deliberately deferred to a second version to keep the first build focused and buildable.

The design and architecture phase — captured in this document — is complete, with every major component selected and justified, the radio and mesh protocols specified, the power budget modeled, and the interaction design resolved. The next phase is a functional two-device prototype built on development boards, intended to validate the full software and protocol stack and produce a working "find your friend" demonstration before any miniaturization or custom hardware. Note: As of this revision that prototype is substantially built: all peripherals are working on dev-board hardware, the navigation stack has been ported to firmware and verified against its original test vectors, and the LoRa link has been field-tested to 635 m through residential obstruction — inside the range this document predicted from its link budget. Remaining work before the two-device demonstration is magnetometer calibration and the mesh protocol port.

The most significant open risk: the real-world RF performance of any such device in a crowd of tens of thousands cannot be known from analysis alone, and will be resolved through staged field testing at progressively larger events. The expectation is that early hardware may perform reliably at smaller-scale festivals first — which is itself a worthwhile target, since the overwhelming majority of events are far smaller than the largest. This document captures both a complete paper design and a clear-eyed account of what building the device will still have to prove. Note: First measurements are encouraging on this point — 635 m through residential obstruction, consistent with the paper estimate. But houses are static where a crowd is not, and a single body was measured to absorb 11.2 dB, so the crowd case remains the open question it always was.

Radio Preset Selection:

Radio Preset	Alt Preset Name	Data-Rate	SF / Symbols	Coding Rate	Bandwidth	Link Budget
Short Range / Turbo	Short Turbo	21.88 kbps	7 / 128	4/5	500 kHz[1]	140dB
Short Range / Fast	Short Fast	10.94 kbps	7 / 128	4/5	250 kHz	143dB
Short Range / Slow	Short Slow	6.25 kbps	8 / 256	4/5	250 kHz	145.5dB
Medium Range / Fast	Medium Fast	3.52 kbps	9 / 512	4/5	250 kHz	148dB
Medium Range / Slow	Medium Slow	1.95 kbps	10 / 1024	4/5	250 kHz	150.5dB
Long Range / Turbo	Long Turbo	1.34 kbps	11 / 2048	4/8	500 kHz	150dB
Long Range / Fast	Long Fast	1.07 kbps	11 / 2048	4/5	250 kHz	153dB
Long Range / Moderate	Long Moderate	0.34 kbps	11 / 2048	4/8	125 kHz	156dB
Long Range / Slow (depr.)	Long Slow	0.18 kbps	12 / 4096	4/8	125 kHz	158.5dB

This choice requires balancing 3 key components: Maximum effective range in a crowded environment, sufficient channel capacity to support multiple devices in a crowded, outdoor environment, and reasonable transmit airtime per packet while accounting for battery capacity.

Long Fast will provide a 153dB link budget, which allows for ~500-800m range in dense crowds (accounting for 20dB of body absorption & multipath loss), At 1.07kbps, and roughly 120bit packets, each packet will occupy the channel for ~110ms, so 3 packets (flooding) will use roughly ~330ms of time. This equals ~33% of channel utilization, plenty of headroom for collisions & possible SOS messaging

Long Moderate provides a 156dB link budget, which would have roughly +40% range but was rejected mostly due to its slow data-rate. The 120 bit packages (x3 = 360 when accounting for flooding) will require 1050ms per message, which will saturate the channel with multiple devices, which will increase the position update intervals to 30+seconds per update.

Medium Slow, at 1.95kbps and roughly -20% range is still under consideration. 2x channel capacity seems reasonable for a 20% range tradeoff, but for now **Long Fast** will be the choice.

Field measurement update. The 500–800 m estimate above was subsequently measured at 635 m through residential obstruction, with SNR operating near the SF11 demodulation floor at the far end — confirming both the link-budget reasoning and the Long Fast selection. See Field Validation for the full data.

Mesh Algorithm:

The Kandi mesh will use a managed flooding system to transmit messages between members of the group. This approach was chosen after learning more about the Meshtastic protocol, because of its simplicity and robustness for smaller networks. With a target group size of <8 users as well as festival grounds of <1km, managed flooding provides enough efficiency and simplicity to avoid dealing with more complex dynamic route learning. There are three main parts to managed flooding to make it work.

Hop Limit:

Every packet comes with a predetermined “hop limit” variable (likely 3), which de-increments by 1 every time it reaches a new node. So a packet with a hop limit of 3 will be retransmitted a maximum of 3 times before it is killed. This prevents a packet from bouncing around the network forever, taking up valuable channel bandwidth.

Duplicate Suppression:

Every packet will have a unique ID, and every receiver will keep a small (~30) buffer of its last received IDs. If a receiver catches a packet with an ID in its buffer, it will kill the message instead of rebroadcasting.

SNR-Weighted Relay Delay

Before relaying, devices will wait a delay that is inversely proportional to its received signal strength, allowing weaker (further) devices to have priority when retransmitting. This allows the packet to be sent further and gives it a better chance of reaching its target destination.

Group Separation will be implemented using a shared-key encryption. During the pairing phase, each group will have a unique 128bit AES key. If a packet fails to decrypt it will be dropped, allowing multiple groups to exist on the same frequency without interference.

Position data will be broadcast every 10 seconds per device, with +- 2 seconds of random delay to prevent collisions.

GNSS Module Selection

The positioning system will use the u-Blox MAX-M10S GNSS module. This chip supports concurrent reception of GPS, GLONASS, Galileo, and BeiDou systems, critical for maintaining a position fix in situations where line-of-sight to a single constellation system is unreliable, such as festivals (bodies, equipment, structures). Its power consumption is rated at ~25mA for initial acquisition and ~12mA for tracking. With aggressive duty cycling, such as a 2 second fix every 10 seconds, effective average current is ~3-5mA.

Its cold-start time is normally 30-150 seconds, which would generally be unacceptable from a user standpoint. This can be mitigated by preloading AssistNow Offline ephemeris data via USB or BLE before users enter the venue, reducing the first fix to <10 seconds. The bracelet will have a companion mobile app used only for initial bonding and A-GNSS data transfer, which will handle all of the above before losing cell service. Note that this creates a condition where users have to pair their devices *before* entering festival grounds or cellular dead zones.

The expected positioning accuracy is 3-5m with open skies and 5-15m in dense crowds with sky obstructions. This is consistent with the reported ~20ft accuracy of existing systems such as CrowdCompass. This would be “good enough” on a macro scale, but it does push for the addition of UWB for close range guidance (detailed in the following section).

Ultra Wide Band

For the first iteration (v1), UWB will not be considered. While a chip like the Qorvo DW3000 provides ~10cm accuracy at <30m range, and would enable a “final approach” UX when the user approaches the target, there are several reasons it will not be included in v1. It adds a significant cost (~$30), requires an additional antenna that would mess with form factor constraints, and handoff between UWB and GNSS adds more complications.

As a temporary fix, the v1 UX could haptically vibrate or visually signal to the user that they are within 10-20m, instructing the user to look for the target visually. UWB will be revisited in v2 once the rest of the core concepts and features are integrated properly.

Microcontroller Selection:

The system will use a Nordic nRF52840 ARM Cortex-M4 microcontroller. Selection criteria, in priority order: (1) ultra-low power consumption including deep-sleep current under 5µA, essential for hitting the 6+ hour battery target (2) integrated Bluetooth Low Energy 5.x with Long Range support, used for the proximity-based bonding before devices switch to LoRa mesh communication (3) sufficient peripheral count to interface with the SX1262 (SPI), MAX-M10S (UART), IMU (I2C), and addressable LED ring (SPI or dedicated PWM) (4) mature community support to minimize firmware development risk.

The nRF52840 satisfies all criteria with significant headroom: 1MB flash and 256KB RAM exceed any plausible firmware footprint, and the chip is supported by both Nordic's nRF Connect SDK (Zephyr-based, production-grade) and Arduino (suitable for rapid prototyping).

Communication interfaces: SX1262 over SPI, MAX-M10S over UART (with PPS pin for precise time synchronization), BNO055 IMU over I2C, WS2812-style addressable LEDs over a dedicated PWM/timer pin. Total pin count fits comfortably within the nRF52840's GPIO availability.

Adaptive duty cycling

Power optimization is governed by GPS-derived stationarity rather than IMU motion alone, to correctly handle festival contexts where users dance in place without locomoting. The device enters stationary mode when GPS displacement stays under 10m for >5 minutes. In stationary mode: GNSS duty-cycles to ~10% (3s active per 30s), position updates transmit at 60-second heartbeats rather than 10s, and the LED ring power-gates fully off between wrist-raise events. The IMU is used in two roles: wrist-raise detection (for LED activation, distinguishable from dancing via forearm rotation pattern), and significant-motion early-wake as a hint to resume normal GPS cadence before the next scheduled wake. This design correctly handles the "stationary at a stage while dancing" case that would defeat naive IMU-triggered approaches.

Power Budget

A component-level power budget was constructed from datasheet current figures and estimated duty cycles for each operating state. Under nominal festival usage, total average current is ~17.5mA, yielding ~22.8 hours of theoretical runtime on a 400mAh LiPo cell (1480mWh). Accounting for real-world inefficiencies not captured in the model — DC-DC converter losses (~10-15%), sleep-state leakage, and wake/sleep transition overhead — realistic runtime is estimated at 13-16 hours, comfortably exceeding the 8-12 hour festival-day target.

The dominant consumer is the GNSS module in continuous tracking mode (9.5mA, ~54% of total budget). This motivates the adaptive duty-cycling strategy described above: dropping GNSS to ~10% duty in stationary mode reduces its contribution to ~1mA, which alone extends runtime by an estimated 30-40%.

The second-largest contributor is the WS2812 LED ring's quiescent draw (~4mA), which persists even when no LEDs are illuminated because the addressable LED ICs consume idle current whenever powered. This is mitigated in v1 by a power-gating MOSFET on the LED supply rail, cutting idle LED current to near-zero between wrist-raise events.

Given the comfortable runtime margin, a 300-400mAh cell (~25×25×4mm) is sufficient, balancing battery life against wrist-worn comfort and enclosure thickness. Final battery sizing will be confirmed against measured current draw on prototype hardware.

Component	State	Current (mA) @ 3V3	Duty Cycle	Avg Current (mA)	Notes
nRF52840 MCU	Sleep	0.00235	90%	0.002115	System ON, full 256 kB RAM retention, wake on any event, power-fail comparator enablec
	Active (@64mhZ)	6.3	10%	0.63	brief wakeup to process packets, update LEDs, some math, button presses...
	Radio Transmitting BLE	16.4	0%	0	Only during bonding, not part of runtime
SX1262 LoRa radio	Sleep	0.0012	86%		
	Transmit @ 14 dBm	25.5	4% (110ms * 3-4 = ~400ms)	1.02	868/915 MHz
	Receive	5.3	10% (RX Duty Cycle Mode)	0.53	DC-DC Mode, Rx Boosted LoRa @ 125kHz
MAX-M10S GNSS	Acquisition	11.5	1% (only at boot)	0.115	GPS+GAL +BDS B1I (default)
	Tracking (Steady State)	9.5	99%	9.405	GPS+GAL +BDS B1I (default)
	Backup	0.028	0% (not for v1)	0	Total current in hardware backup mode V _ BCKP = 3.3 V , V _ IO = VCC = 0 V
BNO055 IMU	Normal Mode	12	5% (active for wrist raised events)	0.6	https:/
	Low Power Mode	0.4	95% (default state)	0.38	
	Suspend	0.04	0%	0	
WS2812 LEDs	Active (2 LEDs lit, 40% brightness)	20	5%	1	8 LEDs total, 1-3 lit at moderate brightness at a time, when wrist raised
	Idle (all off, chips quiescent)	4	95%	3.8	
	Off	0.5	0%	0	
Misc Overhead	Always	1.5	100%	1.5	
			Sum:	17.482115	

Battery = 400mAh
Battery Voltage = 3.7V
Battery Energy = 1480 mWh
Avg Current = 17.5 mA
Avg Power = 64.75mW
Runtime Hours = 22.85 Hours

Field Validation - Phase 2 Measurements

The first field measurements were taken on dev-board hardware (LilyGo T-Beam Supreme, SX1262 at 915MHz, Long Fast preset) in a residential neighborhood with single-family houses in the signal path. Both boards held a 12-satellite fix at HDOP 0.6–1.0 throughout, so reported distances are GPS-derived and accurate to a few meters.

Distance	RSSI (dBm)	SNR (dB)	Segment Delivery
139m	-78	+5.8	-
288m	-113	-8.2	73%
390m	-107	-1.2	88%
635m	-119	-13.5	63%

Packets were still arriving at 635 m, where testing ended. Segment delivery is derived from the change in received-versus-transmitted counters between readings rather than the cumulative ratio, which is distorted by the receiver booting after the transmitter.

SNR, not signal strength, is the binding constraint. The SX1262's raw sensitivity at SF11/250 kHz sits well below the −119 dBm measured at 635 m, so absolute signal level was not the limiting factor. The limit was the demodulation floor: LoRa at SF11 requires roughly −17.5 dB SNR, and far-end readings fluctuated between −13.5 and −18. The link was operating within a few dB of its mathematical floor, which means 635 m approximates the practical maximum in this environment rather than merely the point at which walking stopped.

The link budget prediction holds. This document estimated ~500–800 m of effective range from a 153 dB link budget. The measured 635 m through residential obstruction falls inside that window, validating both the link-budget reasoning and the Long Fast selection — with the caveat that static houses are not a dense crowd, and the crowd case remains untested.

Path loss in this environment is $n \approx 3.7$. Comparing the 200 m and 635 m readings (SNR +5.0 and −13.5), a 3.2× increase in distance cost 18.5 dB, implying a path loss exponent of about 3.7. Free space is $n = 2$; dense urban environments typically run 3–4.

Position freshness degrades before the link does. Past roughly 550 m, the interval between successfully received position updates rose to 15–25 seconds against a 10-second transmit cadence, meaning one or two consecutive packets were routinely lost while the link remained nominally alive. This is the most product-relevant result of the test: for a find-your-friend device, a 20-second-old bearing is materially different from a live one. It provides direct justification for the stale/out-of-range LED state specified in the LED UX section, which was designed before any measurement existed.

Body absorption: 11.2 dB. At a fixed 200 m separation, SNR measured +5.0 dB with the board held clear of the body and −6.2 dB pressed against the chest facing away — 11.2 dB of loss from a single human body. Applying the measured path loss exponent, the 11.3 dB of remaining margin above the SF11 floor in the body-worn case corresponds to roughly a 2× distance multiplier, projecting approximately **410 m body-worn versus 635 m held clear** in the same environment.

Two consequences follow. First, this is direct empirical support for the central radio argument of this document: 915 MHz was chosen over the Totem Compass's 2.4 GHz specifically because water absorbs 2.4 GHz more strongly, and body absorption is now a measured quantity rather than a cited principle. Second, antenna placement relative to the wrist becomes a first-order form-factor constraint rather than a detail — an 11 dB difference between the outward-facing and skin-facing sides of a wristband is a 2× range difference decided by nothing but which surface the antenna occupies.

Magnetometer/LED interference: not detectable. The Open Questions section flagged possible magnetometer disturbance from WS2812 switching currents. Measured with the ring dark and then with multiple LEDs lit at moderate brightness, board held still, no shift was observed. This risk is closed, and the firmware does not require interleaved LED-drive and magnetometer-sampling windows.

Hard-iron distortion characterized. Raw magnetometer readings on the dev board measure ~299 μT total against Earth's ~50 μT field, indicating a hard-iron offset of roughly 250 μT, attributed primarily to the steel springs of the 18650 holder. Because Earth's horizontal component at this latitude is only ~20 μT, this offset predicted heading could swing no more than about ±4.7° through a full rotation. A confirming test found heading trapped within a 9° band across a full 360° rotation. Calibration is therefore a functional prerequisite, not a refinement — and because the offset depends on the battery's presence, it must be performed in the final hardware configuration.

LED UX Design

The interface must communicate direction, distance, and identity for multiple group members simultaneously, remain glance-readable in strobe-heavy low-light conditions, and require minimal interaction from a user who may be impaired, in a dense crowd, and unable to read any on-screen text. These constraints drive a deliberately minimal, light-only interface with no screen.

The device uses an 8-LED ring providing 45° of directional resolution, which is sufficient for "head that way" navigation — humans cannot navigate to finer angular precision while moving through a crowd. Two recessed center buttons handle all input (button 9: pairing/mode control; button 10: SOS).

Three properties are encoded independently so they can be read at a glance:

- **Direction** is encoded by LED *position* — the LED nearest the bearing to the target illuminates.
- **Distance** is encoded by *blink rate* — slow blink indicates farther, faster blink indicates closer. Rate was chosen over brightness because motion is far more perceptible than brightness gradation in chaotic, high-intensity ambient lighting.
- **Identity** is encoded by *color* — each member is assigned a unique color during bonding.

Two colors are reserved system-wide and excluded from member assignment: **yellow** (low battery) and **green** (charging). Reserving these prevents any ambiguity between a member's directional indicator and a system-status signal. In practice the reservations are nearly free, since a device is not actively navigating while charging, and low-battery is a rare state.

Display modes. The interface has two modes to balance ambient group awareness against focused single-target navigation.

Group view is the default resting state. Every bonded member is shown simultaneously, each on the LED corresponding to their bearing, in their assigned color and at a blink rate reflecting their distance. When multiple members fall in the same direction, that single LED cycles through their colors sequentially (e.g. purple–blue–purple–blue), with each member's segment still blinking at the rate corresponding to that member's distance. This mode requires zero interaction and provides continuous awareness of the whole group's rough positions.

Focus mode is entered on demand when the user needs to navigate to one specific person. In focus mode the ring displays only the selected member — single color, single direction, single blink rate — maximizing legibility for the active navigation task. The user double-taps button 9 to enter focus mode, single-taps to cycle through members (all LEDs briefly show the candidate's color as the user cycles), and the selected member is locked as the navigation target. A double-tap returns the device to group view.

This split reflects actual usage: most of the time the user wants ambient awareness of where everyone is, but occasionally needs to actively walk to one person, at which point the display should simplify to exactly that.

State signaling. The full set of display states:

State	Signal
Navigating — far	Directional LED, slow blink, member's color
Navigating — close	Directional LED, fast blink, member's color
Arrived (<10m)	Full-ring flash in member's color (~8s), then a steady single-LED indication; auto-cancels on any button tap. Accompanied by haptic pulse.
Stale / out of mesh range	Directional LED at last-known bearing, dimmed "breathing" pulse to signal the position is no longer live
Multiple members, one direction	Single LED cycles through members' colors, each at its own distance-derived blink rate
SOS received	LED(s) toward the activator's bearing, blinking very rapidly (visibly faster than the close-proximity rate), in the activator's color, with a distinct sustained haptic pattern
Low battery	All LEDs flash yellow briefly on a periodic interval (see priority handling below)
Charging	All LEDs pulse green
Bonding active	Ring chases/spins during pairing

The SOS signal is intentionally kept directional rather than overtaking the entire ring, so the most critical information — *which way the person in distress is* — is preserved. Its blink rate is set faster than any normal navigation state so it is immediately distinguishable, and it is reinforced with a unique haptic pattern so it registers even when the user is not looking at the device.

The arrived state is the product's payoff moment and is given a deliberately satisfying full-ring flash, but with a defined exit condition: it decays to a quiet steady indication after ~8 seconds, or cancels immediately on any button press, so the device does not continue demanding attention once the user has obviously reunited.

Input gesture map. All interaction is layered onto the two buttons via tap-versus-hold and single-versus-double gestures, avoiding additional physical buttons that would increase enclosure complexity and sweat-sealing surface area, and that would be difficult to distinguish by feel in the dark.

Input	Action
Button 9 — hold (3s)	Enter pairing / bonding mode
Button 9 — double-tap	Toggle focus mode on/off
Button 9 — single-tap (in focus mode)	Cycle to next member; selection locks as navigation target
Button 10 — hold (3s)	Trigger SOS broadcast

The pairing button (9) is free to carry the focus-mode functions because pairing occurs once at the start of an event and is not used during normal operation, leaving its tap and double-tap gestures available mid-festival.

State priority hierarchy. Because several states can be true simultaneously (e.g. navigating while the battery is low while an SOS arrives), a strict priority order governs what the ring displays:

1. **SOS (received or self-activated)** — overrides all other displays. Directional, so navigation context is retained.
2. **Arrived celebration** — the brief full-ring flash on reaching a target.
3. **Focus-mode single-target display** — when the user has actively selected one member.
4. **Group-view default** — ambient display of all members.
5. **System status (low battery)** — rather than permanently occupying the ring and obscuring navigation, low-battery interrupts with a brief all-LED yellow flash on a periodic interval (e.g. once per 30–60s), returning to the navigation display between flashes. (Charging does not contend, as the device is docked and not navigating.)

This hierarchy ensures that the most safety-critical information is never suppressed by lower-priority status signals, while still surfacing those signals reliably.

Bonding Flow

Bonding is the process by which a set of devices establishes a shared group identity before an event: a common encryption key (for group separation on the shared LoRa channel), a per-member color assignment, a designated leader, and — where possible — preloaded A-GNSS data. The interaction is constrained by the same screenless, low-input philosophy as the rest of the device, but unlike normal operation, bonding occurs *before* entering the venue while cell service and a companion phone are typically available. The design exploits this by using a different radio and a different interaction model for bonding than for runtime.

Why BLE, not LoRa. Bonding uses the nRF52840's onboard Bluetooth Low Energy rather than the LoRa mesh. BLE is appropriate here for several reasons: it is short-range by nature, which provides inherent proximity-gating (you can only bond with devices physically near you); it supports an authenticated, encrypted link via LE Secure Connections, which LoRa does not provide at the link layer; and it is the natural transport for communicating with the companion phone app. LoRa is reserved exclusively for the in-venue mesh, where its range and penetration matter; bonding has no need for range and benefits from BLE's pairing security.

Primary path: app-mediated bonding. The expected setup flow uses a companion mobile app, which is required regardless for A-GNSS preloading. The group leader creates a group in the app; the app generates a random 128-bit AES key and assigns each member a color from the available palette (excluding the reserved yellow and green). Each member's device is paired to that member's phone over a BLE Secure Connections link, and the app provisions each device with the group key, the member's own ID and color, the full roster of other members' IDs and colors, the leader's ID, and current A-GNSS ephemeris data. Because the AES key is delivered over the encrypted, authenticated BLE link established by the phone pairing, it is never transmitted in the clear. The app screen makes the otherwise-awkward parts of a screenless device trivial — seeing who is in the group, confirming color assignments, and designating the leader.

Fallback path: device-to-device proximity bonding. For cases where the app or cell service is unavailable, devices can bond directly. The leader holds button 9 (3s) to enter group-creation mode, during which the device generates the group key and broadcasts a BLE advertisement; the LED ring chases to indicate active bonding. Each joining member holds button 9 on their own device to scan for a nearby broadcasting device. Joining is proximity-gated by BLE signal strength — the devices must be within roughly one to two meters (effectively, held together) for a join to be accepted, which prevents accidental cross-joining with other nearby Kandi groups. The two devices establish an LE Secure Connections link (ECDH key exchange, so a passive eavesdropper cannot derive the link key), over which the group key and a color assignment are transferred. Colors are auto-assigned in join order from the reserved-excluded palette, and the initiating device is the leader by default. A confirmation flash on both devices indicates a successful join.

This fallback is functional but carries two limitations relative to the app path: it cannot preload A-GNSS data (so first-fix on entry will be slower), and because the screenless devices cannot perform the passkey or numeric-comparison step that normally defends against an active man-in-the-middle, its security rests on proximity-gating and a short bonding window rather than full cryptographic MITM resistance. For the threat model of a consumer festival accessory, this is an acceptable tradeoff; it is noted here as a known constraint rather than a solved problem.

What is provisioned to each device. On completion of bonding, each device persistently stores: the 128-bit group AES key; its own member ID and color; the roster of other members (ID → color), which is what allows incoming position packets to be mapped to the correct directional color at runtime; the leader's ID, for the reserved designator treatment; and (app path only) A-GNSS ephemeris data, which has a limited validity window of hours to a few days.

Timing constraint. The full setup — group creation, key and color distribution, and especially the A-GNSS preload — depends on internet connectivity and should therefore be completed before entering the venue or any cellular dead zone. This imposes a real usage requirement: groups must bond ahead of time, not after networks have already saturated. This constraint is surfaced to the user through the companion app and is the single most important piece of pre-event guidance, since a group that fails to bond beforehand cannot establish a shared key once inside without falling back to the slower, app-less device-to-device path (which still requires the members to be physically together).

LED feedback during bonding. Throughout bonding the ring displays a chasing/spinning animation to indicate the device is in an active bonding state and is not yet operational for navigation. A successful join or provisioning is confirmed by a brief flash in the member's newly assigned color, giving immediate, screen-free confirmation of both that bonding succeeded and which color the member has been given.

Form Factor

The physical design is the area with the highest residual uncertainty in this document. Unlike the radio, mesh, and power decisions — which can be reasoned about from datasheets and protocol specs — antenna performance, fit, comfort, and sealing behave in ways that can only be validated with physical prototypes. The decisions below are therefore design intentions and starting points, explicitly subject to revision once hardware exists.

Layout. The device is built around a ~40mm circular face. The 8-LED ring occupies the perimeter of the face, because its entire function is directional — each LED must map to a compass bearing — so it has to sit at the outer edge where angular position is meaningful. The two control buttons sit in the center, recessed. Below the face, the main PCB carries the MCU, both radios, the GNSS module, the IMU, and the LED driver. The LiPo cell sits beneath the PCB and is the dominant driver of overall thickness. The realistic prototype thickness is on the order of 12–15mm — thicker than a production wearable would target — because prototyping prioritizes accessible, hand-assemblable construction over slimness.

Antenna strategy. This is the central physical-design challenge. The device contains three radios with conflicting antenna requirements, all to be packaged into a small puck worn on a wrist — which is among the worst possible RF locations, being adjacent to a large mass of RF-absorbing tissue and frequently oriented pointing at the ground.

The *GNSS antenna* needs a clear view of the sky. A ceramic patch antenna mounted on the top of the PCB, under the non-metallic face, is the realistic approach. This placement has a fortunate property: when the user raises their wrist to look at the device, the face — and the patch — points roughly upward, giving the best sky view at exactly the moment the user is actively navigating. Degraded performance when the wrist is lowered is accepted; position simply updates on the next good fix.

The *sub-GHz LoRa antenna* is the hardest. A quarter-wave antenna at 915MHz is ~8.2cm, far too long to fit inside a 40mm enclosure. The conventional fallbacks — a chip antenna or a meandered PCB trace antenna — fit, but sacrifice 6–10dB of efficiency, which directly cuts the range that the entire product depends on. A more promising option specific to a wrist device is to route the LoRa antenna into the *wristband itself*: the silicone strap is long enough to house a flexible antenna trace approaching a usable fraction of a wavelength, recovering antenna efficiency that a face-confined antenna cannot. This is proposed as the primary approach to evaluate, with a chip/PCB antenna as the fallback if band integration proves impractical. Either way, real-world range must be measured, not assumed.

The *BLE 2.4GHz antenna* is the easiest and lowest-priority of the three — a small chip antenna or PCB trace on the main board is sufficient, since BLE is used only for short-range bonding before the event, not for in-venue range.

Measured body absorption constrains placement. Field testing found 11.2 dB of loss between the board held clear of the body and pressed against the chest, projecting to roughly a

2× range difference. Antenna placement relative to the wearer's arm is therefore a first-order constraint on this design. This complicates the band-integrated proposal above: a strap wraps the wrist, so any antenna routed through it necessarily has sections against skin. The radiating section should be biased toward the outer surface where possible, and the efficiency cost of doing so must be measured rather than assumed.

One property of the existing interaction model partially mitigates this. The user raises the wrist to read the LED ring, which simultaneously lifts the antenna away from the body and rotates the face skyward — improving both the LoRa link and the GNSS sky view at precisely the moment the user is actively navigating. Position freshness is best exactly when it matters most. This is an emergent benefit rather than a designed one, but it is worth preserving in any subsequent form-factor revision.

Magnetometer placement. The directional UX depends on the IMU's magnetometer to know which way the wrist is facing (without it, the LED arrow only works at a fixed wrist orientation — the exact limitation of the Totem Compass). Magnetometers are sensitive to nearby current loops and ferrous material, both of which are abundant in this design: the LED ring's switching currents and the battery's charge/discharge currents are close by in a tightly packed enclosure. The IMU should be placed as far as practical from the LED current traces and the battery, and the firmware must implement hard-iron calibration (and ideally a user-facing recalibration gesture, as existing products require). Dynamic interference from the LEDs switching during use is a known risk for this class of device and is flagged for prototype validation — it may require driving the LEDs and sampling the magnetometer in interleaved time windows.

Charging and sealing. The target is IP67 sweat and water resistance, which is in tension with the USB-C port specified in the original concept. Every opening and moving part — the port and the two buttons — is a sealing liability. The recessed buttons can be sealed with membranes or gaskets. The USB-C port is the harder problem; a gasketed port is possible but is the most likely point of ingress failure. For this reason, contact-pin (pogo) charging or Qi wireless charging are worth evaluating as alternatives, because both allow a fully or near-fully sealed enclosure and materially improve achievable IP rating for a device that will be soaked in sweat across an 8-hour festival day. USB-C remains the baseline for prototype convenience, but charging method is an open decision pending the sealing evaluation.

Comfort and band integration. The device must remain comfortable for 8+ hour wear. Weight is dominated by the battery and PCB; a soft silicone strap distributes load and tolerates sweat better than rigid materials. The kandi aesthetic is achieved not through injection-molded imitation but through genuine kandi construction — pony beads on elastic cord surrounding or accenting the silicone band — which is authentic to the culture, since kandi is inherently handmade. This also means the decorative layer is the one part of the device that is deliberately *not* engineered to tight tolerance: it is hand-assembled, variable, and personal, which is the point.

Prototype versus production. The form factor described here is for a functional prototype, optimized for buildability and iteration rather than for the slimness, cost, or manufacturability of a finished product. Several decisions — final thickness, antenna implementation, charging method, sealing approach — are expected to change as physical units are built and measured. This section captures the starting design and, as importantly, the specific physical risks that prototyping must resolve.

Open Questions and Risks

This section consolidates the unresolved questions and known risks identified throughout the design. They are grouped by area, and each notes how it is expected to be resolved. The intent is to be explicit about what this paper design does *not yet* know, since the gap between a sound design on paper and a working device in a 140,000-person crowd is precisely where products like this succeed or fail.

RF environment and range (highest risk). This is the single largest source of uncertainty and the most likely to invalidate core assumptions. A large festival is among the most hostile RF environments that exists: tens of thousands of phones saturating the 2.4GHz band, dense crowds of RF-absorbing bodies, multipath reflection from metal stage and scaffolding structures, broadband EMI from lighting and laser rigs, and a growing population of other sub-GHz devices. The choice of sub-GHz LoRa is specifically intended to mitigate the worst of this (body penetration, avoiding the 2.4GHz roar), but the actual effective range in a packed crowd — estimated here at ~500–800m but plausibly much less — cannot be known without field testing. A bench or open-park test will dramatically overstate real performance. The resolution path is staged field testing at progressively larger events (a park, then a small 1–5k event, then larger), with the explicit expectation that v1 may only perform reliably at sub-EDC scale, which is still useful since most festivals are far smaller than EDC.

A related open question is channel capacity at scale. The capacity analysis assumes a single 8-device group, but a real festival could host hundreds of independent Kandi groups sharing the same frequency. Group separation via encryption keys handles *interference at the application layer* (foreign packets are dropped), but every group's traffic still consumes shared airtime. Whether the channel saturates with many concurrent groups is unvalidated and may require adaptive measures (lower update rates under congestion, listen-before-talk).

Append: *Status: partially validated.* Field testing measured 635 m through residential obstruction, inside this document's 500–800 m estimate, with SNR within a few dB of the SF11 demodulation floor at the far end. This confirms the link-budget reasoning but does not resolve the crowd case: houses are static obstructions, whereas a festival adds hundreds of RF-absorbing bodies, a substantially higher noise floor, and continuously changing multipath geometry. The staged escalation to progressively larger events remains the resolution path.

Separately, the measured 11.2 dB of absorption from a single body suggests this document's 20 dB allowance for combined body absorption and multipath in a crowd is more likely optimistic than pessimistic. Absorption does not accumulate linearly per person — RF diffracts and reflects around obstructions — so no direct extrapolation is claimed, but the 20 dB figure should be treated as a floor rather than an estimate.

Antenna performance. The proposal to route the LoRa antenna into the wristband is promising but unproven; if it proves impractical, the fallback chip/PCB antenna sacrifices 6–10dB and correspondingly reduces the range the product depends on. GNSS antenna performance under real wrist-orientation conditions is similarly unknown. Both resolve only by building and measuring physical units.

Append: Body absorption is now measured at 11.2 dB, making antenna orientation relative to the wrist a quantified first-order constraint. The band-integrated proposal must be evaluated against this figure, since any strap-routed antenna necessarily has sections against skin.

Positioning and navigation accuracy. GPS accuracy in dense crowds (estimated 5–15m) may or may not be sufficient for the close-range "find my friend" experience without UWB. The deferred-UWB decision rests on the assumption that a haptic "you're close, look around" cue compensates adequately; whether the arrived experience actually feels satisfying rather than clunky is a UX question only user testing can answer, and may force UWB earlier than v2. Separately, the GPS-displacement-based stationary detection — the basis for the adaptive power savings — assumes GPS jitter can be reliably distinguished from real locomotion via a 10m threshold; in practice, crowd multipath could inflate jitter and cause false transitions in either direction. This needs validation against real GPS traces in a crowd.

Append: Field testing surfaced a freshness issue independent of accuracy: past ~550 m, intervals between received position updates reached 15–25 seconds against a 10-second transmit cadence. Bearing staleness, not only positional error, degrades the navigation experience at range, making the stale-state LED indication load-bearing rather than cosmetic.

LED UX legibility. Several UX assumptions are untested under real conditions. Whether 45° (8-LED) angular resolution feels sufficient for navigation, whether group-view remains readable when displaying up to seven colors at once or instead becomes visual noise in strobes, and whether blink-rate distance encoding is perceptible in chaotic high-intensity lighting are all open. These resolve through user testing in representative lighting, and may drive changes to LED count, the group-view/focus-mode balance, or the encoding scheme. SOS false-activation is a specific risk: a single accidental SOS erodes group trust in the feature, so the 3s-hold gesture and its haptic confirmation need validation that they are both hard to trigger accidentally and reliable under stress.

Hardware and physical integration. Magnetometer interference from LED switching currents has been tested on dev-board hardware and was not detectable; this risk is closed and interleaved LED/magnetometer sampling is not required. Interference from battery currents in a tightly packed final enclosure remains untested. Hard-iron distortion has been characterized at ~250 μT against a ~20 μT horizontal signal, confirming calibration as a functional prerequisite. IP67 sealing is in tension with the USB-C port and button penetrations, and the charging method (USB-C vs pogo-pin vs wireless) remains an open decision pending a sealing evaluation. Thermal behavior against the wrist and comfort over 8+ hours are unvalidated. All of these resolve only with physical prototypes.

Power. The power budget is constructed from datasheet figures and estimated duty cycles and is therefore optimistic; real runtime is expected at roughly 60–70% of the theoretical ~22.8 hours, and the projected 30–50% gain from adaptive duty cycling is a model, not a measurement. Actual consumption — including DC-DC losses, sleep-state leakage, and wake/sleep transition overhead — must be measured on a bench with a current meter before battery sizing is finalized. The LED quiescent-draw mitigation (a power-gating MOSFET) likewise needs to be confirmed effective in hardware.

Security. The screenless device-to-device bonding fallback cannot perform the passkey or numeric-comparison step that defends against an active man-in-the-middle during BLE pairing, and relies instead on proximity-gating and a short bonding window. This is judged acceptable for a consumer festival accessory but is not cryptographically complete. More broadly, the LoRa mesh's resistance to adversarial behavior (replay, spoofing, jamming) beyond basic duplicate suppression has not been analyzed and is out of scope for v1.

Operational constraint. The dependence on pre-event bonding and A-GNSS preloading imposes a hard usage requirement: groups must complete setup before networks saturate. A group that fails to do so cannot establish a shared key in-venue except via the slower, app-less device-to-device path, which still requires members to be physically together. This is a genuine product limitation, not just an implementation detail, and the companion app's most important job is making this requirement unmissable.

Commercialization and regulatory (out of scope for prototype, noted for completeness). A shipping product operating in the 915MHz ISM band requires FCC Part 15 certification (and CE for international sale), a testing process on the order of $15–40k. The original sub-$80 retail target sits against a BOM that — before certification, distribution, returns, and margin — is already tight, and competitor pricing suggests the category may run thin or negative hardware margins. Finally, the product's differentiation (screenless sub-GHz mesh in a Kandi form factor) could narrow if Apple expands offline phone-to-phone finding in future iOS releases. None of these affect the prototype, but they bound the path from prototype to product.

Roadmap

The development is staged so that the riskiest assumptions are tested as early and as cheaply as possible, and so that each phase produces a demonstrable artifact. The guiding principle is to prove the entire software and protocol stack on convenient, oversized hardware *before* committing to miniaturization or custom hardware — going straight to small custom boards without first validating the firmware, mesh protocol, and navigation math is the most common way projects of this kind stall.

Phase 1 — Design & Architecture (complete). This design document. It resolves the component selection, radio and mesh-protocol design, power budget, UX, bonding flow, and physical-design intentions, and inventories the open risks. The document itself is the deliverable and serves as a standalone portfolio artifact independent of any hardware.

Phase 2 — Dev-board functional prototype (primary summer goal). Status: in progress. Peripheral bring-up is complete on the LilyGo T-Beam Supreme platform — PMU rail management, OLED, WS2812 ring, GNSS with live fix, IMU, and magnetometer. The navigation stack (haversine distance and bearing, tilt-compensated compass, bearing-to-LED mapping) has been ported from the Python reference implementation to C++ and passes 19 boot-time self-tests against the original Python golden values. A LoRa link between two boards is operational and has been range-tested in the field. Remaining: magnetometer hard-iron calibration and porting the mesh protocol to firmware.

Platform note: the dev boards use an ESP32-S3 rather than the nRF52840 specified earlier in this document. This is deliberate — Phase 2 validates the software and protocol stack, for which the MCU is interchangeable. The nRF52840's power advantages matter from Phase 3 onward.

Phase 3 — Integration & miniaturization (likely extends past summer). Migrate from dev boards to discrete small components — a compact nRF52840 module (e.g. a Seeed XIAO) with separate SX1262, MAX-M10S, and IMU — assembled on a custom or simple board and running from a LiPo cell with proper power management (charging IC, regulation). This is where the theoretical power budget gets replaced with bench measurements from a current meter, and where the adaptive-duty-cycling and LED-power-gating strategies are confirmed in hardware. Deliverable: a single integrated, battery-powered unit (still in a project enclosure, not yet wrist-worn) with measured rather than estimated power figures.

Phase 4 — Form factor & custom hardware. A custom PCB designed for the wrist enclosure, the antenna implementation (including evaluating the band-integrated LoRa antenna against the chip-antenna fallback), iterative 3D-printed enclosure design, LED diffusion, the sealing/charging-method decision, and kandi band integration. This is the phase that resolves the antenna and physical-integration risks, all of which require physical units to settle. Deliverable: a wrist-worn prototype.

Phase 5 — Scaled field validation. The phase that confronts the existential RF risk. Test at progressively larger real events — beyond the park, to small 1–5k gatherings, then larger — gathering real range and channel-capacity data and iterating on radio settings and protocol behavior accordingly. The explicit expectation is that early units may perform reliably only at sub-EDC scale, which is still a useful product.